

When Regimes Do Not Help: Heavy Tails, Covariance Memory, and Short-Horizon Equity Risk Forecasting

Saverio Lauriola

Working paper — companion to the reproducible notebook
equity-risk-modelling-regime-aware-pca

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Abstract

We study one-day-ahead density and tail-risk forecasting for a panel of US large-cap equities and ask which modelling ingredients survive strict out-of-sample validation. Three deliver robust gains, each established against a strong benchmark: (i) a multivariate Student- t likelihood with estimated tail index $\hat{\nu} \approx 5$, which dominates every covariance refinement by roughly an order of magnitude and corrects expected-shortfall calibration; (ii) a long-memory shrinkage EWMA covariance update with $\lambda \approx 0.9925$, the best covariance model in a fine grid; and (iii) a reactive portfolio-variance layer (fast EWMA or GARCH) that restores Value-at-Risk conditional coverage where static specifications fail. Against these, a battery of regime-aware models—VIX/HMM covariance mixtures, eigenstructure-defined regimes, adaptive spectral forgetting, and a Kalman-filtered factor-premia overlay—does not improve the final out-of-sample result. A direct bootstrap control is decisive: the apparent edge of adaptive eigenstructure forgetting is fully explained by covariance-memory length, not by regime adaptation. A Marchenko–Pastur diagnostic explains why—on this panel a short estimation window resolves essentially one signal mode, leaving correlation-rotation detectors little to act on. The contribution is a model-validation result: the robust short-horizon signal lies in tail shape and covariance memory rather than in regime conditioning or expected-return dynamics, and a candidate model should be judged against the simplest benchmark that could explain its edge.

1 Research question

The project asks whether increasingly sophisticated regime-aware equity risk models improve short-horizon out-of-sample density forecasts *once they are compared against fair covariance-memory and tail-shape benchmarks*.

The initial idea was intuitive: market stress should change the covariance structure of equities, and a static covariance matrix should miss that time variation. The first version therefore compared a global low-rank PCA covariance model against regime-aware alternatives driven by a VIX-based state variable. Later extensions made the test stricter by adding heavy-tailed likelihoods, long-memory EWMA covariance benchmarks, eigenstructure-defined regimes, adaptive-memory covariance models, a Kalman-filtered expected-return overlay, and VaR/ES backtests with exceedance-clustering diagnostics. The final question is therefore not whether a regime model can beat a static baseline, but whether regime conditioning remains useful after controlling for the two effects risk models most often need: tail thickness and covariance memory.

The headline result is summarised in Table 1. The effect hierarchy is explicit and is the organising principle of the paper:

$$\underbrace{\text{tail shape}}_{\approx 2.3\text{--}2.5 \text{ NLL}} \gg \underbrace{\text{covariance memory}}_{\approx 0.6 \text{ NLL}} \gg \underbrace{\text{regime conditioning}}_{\approx 0 \text{ after controls}} \approx \underbrace{\text{expected-return dynamics}}_{\approx 0 \text{ at 1 day}}.$$

Ingredient	Role	Survives OOS?	Decisive evidence
Student- t likelihood ($\hat{\nu} \approx 5$)	marginal / tail shape, ES	Yes (dominant)	$\Delta\text{NLL} \approx -2.3$ to -2.5 vs Gaussian; ES ratio ≈ 0.95
Long-memory shrinkage EWMA ($\lambda \approx 0.9925$)	one-day covariance	Yes (best cov.)	$\Delta\text{NLL} \approx -0.593$ vs global; interior optimum in fine grid
Reactive portfolio variance (EWMA-vol / GARCH)	VaR conditional coverage	Yes	restores Christoffersen cc at 2.5% ($p_{cc} : 0.003 \rightarrow 0.94$)
VIX/HMM covariance mixture	regime covariance	No (not live)	static $P(\text{win}) = 1.00$ but walk-forward $P \approx 0.13$
Eigenstructure regimes / adaptive forgetting	regime covariance	No (con-found)	$P(\text{beats fixed } \lambda=0.99) = 0.000$; edge is memory length
Kalman factor-premia overlay	expected-return layer	No (at 1 day)	$\Delta\text{NLL} \approx +0.001$, $P(\text{beat}) = 0.215$

Table 1: What survives strict out-of-sample validation. Negative ΔNLL means improvement over the stated baseline. Absolute NLL levels are comparable only *within* a section (same constants and window), never across sections.

2 Data and forecasting design

The empirical panel contains daily adjusted-close returns for 24 US large-cap equities, with VIX used as the external volatility/regime proxy. Data are requested from 2010 onward, but the common sample begins in 2012 because the analysis uses the intersection of all available histories; the main test window begins in 2022.

The forecasting design is deliberately causal: models are estimated on training or trailing windows only; state variables and filtered probabilities are lagged before scoring returns; walk-forward sections refit periodically using only past data; the primary score is out-of-sample negative log-likelihood (NLL); and a moving-block bootstrap assesses loss-difference robustness. The target object is the one-day predictive distribution of the return vector, so the modelling effort is directed at the covariance matrix and the likelihood shape rather than at expected-return forecasting.

Reproducibility. All results are produced by the companion notebook on a fixed data vintage; estimation windows, lags, bootstrap block lengths and the ν grid are specified there. Absolute NLL levels differ between the up-to-constant scoring of Section 5.4 (window-level, ≈ -175) and the full-constant five-year-window scoring of the covariance grid (≈ -70); only within-section differences are interpreted.

3 Model classes

Static low-rank PCA (baseline). A low-rank PCA covariance model with $k = 3$ factors, combined with shrinkage for numerical stability. It captures the dominant common equity-risk structure while avoiding a fully unconstrained covariance estimate.

VIX/HMM covariance mixtures. A past-looking VIX state variable (log-VIX with EWMA standardisation) drives a two-state Gaussian HMM, producing filtered calm/stress probabilities p_{t-1} that, with a one-day lag, mix calm and stress covariance matrices:

$$\Sigma_{t|t-1} = (1 - p_{t-1}) \Sigma_{\text{calm}} + p_{t-1} \Sigma_{\text{stress}}. \quad (1)$$

This is not an endogenous latent-state return model; it is an externally driven, interpretable stress-conditioned covariance model. Simpler regime variants—HMM-informed piecewise PCA thresholds, vol-mix/correlation-fixed forms, and hard low/mid/high splits—separate whether any gain comes from probabilistic smoothing, covariance scaling, or sample splitting.

Eigenstructure-defined regimes. The state is replaced by measures derived from the correlation eigenstructure: the absorption ratio (top- k share of correlation variance), PCA-subspace rotation between short and long windows, eigenrank buckets, and adaptive forgetting driven by spectral rotation. The purpose is to test whether regimes defined directly from the covariance object forecast better than external VIX-based regimes.

EWMA covariance benchmark. The key benchmark is a shrinkage EWMA update

$$\Sigma_t = \lambda \Sigma_{t-1} + (1 - \lambda) r_{t-1} r_{t-1}^\top, \quad (2)$$

shrunk toward a stable trailing-window anchor. A wide/fine grid tests $\lambda \in [0.90, 0.999]$.

Student- t likelihood. The Gaussian likelihood is replaced by a multivariate Student- t . The scale matrix is adjusted so the covariance target remains comparable to the Gaussian one, $\text{Cov}(r_t) = \Sigma_t$ (i.e. the scale equals $\Sigma_t(\nu - 2)/\nu$). The tail index ν is estimated by profile likelihood on each trailing window, with empirical median $\hat{\nu} \approx 5$.

Kalman-filtered factor premia. To test whether an expected-return layer adds value, a Kalman filter is applied to the first $k = 3$ PCA factor returns; the filtered premia are mapped back to asset-level means and scored against the same covariance/tail model.

Reactive portfolio-variance layer. The multivariate density forecasts are converted to equal-weight portfolio VaR and ES, and exceedance clustering is attacked with reactive conditional-variance models (fast EWMA-vol and GARCH(1,1)-style portfolio variance).

4 Evaluation metrics

The primary metric is mean out-of-sample NLL, $\text{NLL}_t = -\log f(r_t \mid \mu_t, \Sigma_t)$; lower means a higher predictive density on the realised vector. Model comparisons use the loss difference $\Delta\text{NLL} = \text{NLL}_{\text{model}} - \text{NLL}_{\text{baseline}}$ (negative is better). The risk capstone uses one-day VaR and ES at the 1% and 2.5% levels, evaluated through hit rates, Kupiec unconditional coverage, Christoffersen independence and conditional coverage, and the ES realised-loss-to-forecast ratio on exceedance days.

5 Results

5.1 Heavy tails are the dominant ingredient

Replacing the Gaussian with a Student- t likelihood produces the largest and most stable gain. Across the main walk-forward grid the estimated tail index has median $\hat{\nu} \approx 5$, and the Student- t likelihood improves NLL by roughly 2.3–2.5 points on the main models—an order of magnitude larger than most covariance-model differences. For one-day US large-cap equity density forecasting, tail shape matters more than regime sophistication.

5.2 Long-memory EWMA is the best covariance model

In a fixed- λ grid the long-memory EWMA benchmark beats the static global covariance and the regime/eigenstructure variants; a wider/finer grid confirms this is not a boundary artefact, with an interior optimum near $\lambda = 0.9925$ (Table 2). The covariance conclusion is therefore not “use a reactive regime switch” but “use a stable, long-memory update with mild recency weighting.”

Model	Student- t NLL	Δ vs global
EWMA $\lambda = 0.9925$	−70.780	−0.593
EWMA $\lambda = 0.9900$	−70.763	−0.576
Validation-selected EWMA	−70.754	−0.567
Global	−70.187	0.000

Table 2: Fine-grid covariance comparison (full-constant, five-year window). Negative Δ is improvement.

5.3 Reactive variance restores VaR conditional coverage

The VaR/ES capstone gives the density result a market-risk reading. With the final EWMA covariance, Student- t tails improve 1% tail calibration and ES severity relative to Gaussian tails: the 1% Student- t VaR hit rate is close to target and the ES loss-to-forecast ratio is near one (about 0.95, against 1.1–1.2 under Gaussian tails). At the 2.5% level a sharper picture emerges: exceedances *cluster*, and both Gaussian and Student- t static specifications fail Christoffersen conditional coverage—this is a variance-dynamics problem, not a tail-shape problem. A reactive conditional-variance layer fixes it under either likelihood (Table 3). Tail shape and reactive variance are thus two *orthogonal* interventions: the former controls tail magnitude (ES), the latter controls clustering (conditional coverage).

Portfolio-variance model (2.5%, Student- t)	Christoffersen p_{cc}
Static 5y window	0.003 (fails)
GARCH(1,1)	0.14
Reactive EWMA-vol ($\lambda = 0.94$)	0.46
Reactive EWMA-vol ($\lambda = 0.97$)	0.94

Table 3: Conditional-coverage test at the 2.5% level. Reactive variance declusters exceedances where the static specification fails; power is limited by the exceedance count, so the robust signal is static-vs-reactive, not the ranking among reactive models.

5.4 The regime models, as controlled nulls

The VIX/HMM mixture works statically but not live. In static out-of-sample scoring the HMM calm/stress mixture improves materially on the global baseline ($\Delta\text{NLL} = -0.370$; moving-block bootstrap $P(\text{win}) = 1.00$). But walk-forward recalibration changes the interpretation: the probability that the rolling HMM beats the rolling global benchmark is only about 0.13. The static edge is evidence of useful regime structure on a fixed split, not of a deployable real-time advantage.

Eigenstructure adaptation is a memory-length confound. The adaptive eigenstructure model beats the global baseline and therefore looks attractive. A direct bootstrap control against

the best fixed- λ EWMA dissolves the result:

$$P(\text{adaptive eigen-}\lambda \text{ beats fixed } \lambda=0.99) = 0.000, \quad P(\text{beats fixed } \lambda=0.97) = 1.000, \quad P(\text{beats global}) = 1.000.$$

The adaptive model beats only weaker, faster-memory benchmarks, because it spends most of its time near a slow setting; against a plain long-memory EWMA it loses. The driver is covariance-memory length, not regime adaptation. This single control is the methodological core of the paper.

Why eigenstructure regimes are weak here. A Marchenko–Pastur diagnostic shows that a short detector window contains essentially one reliable signal mode—the market mode. In a panel dominated by a stable market factor, RMT-cleaned rotation detectors are blind to weaker sector/style rotations, so eigenstructure regime conditioning has little to act on. The claim is not that eigenstructure regimes are useless in general, but that this one-day US large-cap panel lacks enough robust covariance-shape rotation for them to beat a simpler long-memory model.

Kalman expected returns add no one-day density value. The Kalman overlay tests whether a low-dimensional expected-return layer helps beyond the final covariance/tail model. It does not: $\Delta\text{NLL} \approx +0.001$ with $P(\text{Kalman beats base}) \approx 0.21$ on the final EWMA covariances. At one day, expected-return dynamics are too weak relative to covariance and tail effects. This does not rule out factor-premia modelling for multi-day allocation; it shows the density gains here are not coming from a mean forecast.

6 Interpretation

The main lesson is negative in a useful way. Several regime models are plausible *ex ante*, and one produces a strong static improvement; but once the tests are made stricter—walk-forward refitting, fair EWMA benchmarks, Student-*t* likelihoods, and direct confound controls—the robust contribution comes from three simpler, *orthogonal* components: a heavy-tailed likelihood (tail magnitude and ES), a long-memory shrinkage EWMA covariance (one-day density), and a reactive portfolio-variance layer (VaR conditional coverage). These compose into the final model; the regime and expected-return channels are the well-documented nulls that justify *not* adding them.

This is a model-validation result. It argues against over-interpreting static regime wins and in favour of asking whether a candidate survives the simplest strong benchmark that could explain its edge. For equity quantitative research it is informative—one-day density quality on this panel is driven more by risk dynamics than by expected-return dynamics—and for market-risk and risk-strategy use it is directly relevant: the final model improves density scoring and tail calibration while avoiding fragile regime overfitting.

7 Limitations and next steps

This is an empirical research note, not a general theorem about regime models. The universe is a survivorship-selected panel of current US large-cap equities; the horizon is one day; the test path starts in 2022; and the result concerns density and tail-risk scoring, not portfolio alpha. Regime conditioning may matter more in multi-asset universes, at multi-day horizons, or under allocation objectives.

The natural next steps sharpen exactly the boundary this paper draws. A *cross-asset* panel (equities, bonds, commodities, FX) introduces a covariance change this equity panel lacks—a sign-flipping stock–bond correlation—and tests the prediction that regime conditioning pays when the between-regime covariance change is large; a unifying statement is that realised regime value scales as the product of the size of the covariance change and the quality of the regime

classifier. A *multi-horizon* extension (HAR-style realised-variance forecasting at 5, 10, 22 days) tests the prediction that the covariance signal strengthens with horizon, in contrast to the one-day null for expected returns. Together these would turn the present single-panel result into a generalisable account of *when* regime conditioning helps; that is the intended journal-grade synthesis, of which this note is the seed.

8 Conclusion

The answer is not that regimes never help. In this one-day US large-cap equity density-forecasting setting, regime complexity does not survive the strongest simple controls. The robust model is a Student- t likelihood combined with a long-memory shrinkage EWMA covariance, with a reactive portfolio-variance layer for VaR conditional coverage. VIX/HMM mixtures, eigenstructure regimes, adaptive forgetting, and Kalman-filtered one-day factor premia are useful tests, but they do not improve the final out-of-sample result once tail shape and covariance memory are properly modelled. The contribution is therefore less a “regime model wins” story and more a model-validation one: the useful result is not the most complex model, but the process of identifying the simpler mechanism that actually survives out of sample.

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